

STINGRAY: A TRANSPORT-CONSTRAINED MESSAGING ARCHITECTURE FOR UNTRUSTED MOBILE ENVIRONMENTS

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ABSTRACT

Modern secure messaging systems provide strong application-layer confidentiality, but many continue to rely on carrier-routed mobile transport as an assumed-safe communications substrate. That assumption is increasingly weak in environments where the cellular signaling and radio path may itself be part of the threat model.

This paper presents Stingray, a transport-constrained messaging architecture designed to reduce exposure to radio-layer interception, carrier-path metadata leakage, and relay-side visibility by removing the cellular radio from the trusted transport set entirely. The system enforces a fail-closed transport discipline, applies end-to-end cryptography above non-cellular transport only, stores relay-visible data as opaque ciphertext, and keeps identity material and message history local to the endpoint.

The central claim of this paper is architectural rather than cryptographic: if the transport path cannot be assumed trustworthy, application-layer encryption alone is not a complete answer. In such cases, reducing trust in the transport stack is preferable to compensating for it exclusively at higher layers.

This paper defines the threat model, explains the layered architecture, outlines the intended security properties, compares the model against conventional secure messengers, and identifies the system's operational limitations and non-goals. Stingray is presented as a research system in alpha state and is published for technical review, threat-model challenge, and architectural scrutiny rather than as a production-ready safety claim.

1. INTRODUCTION

The dominant secure messaging model on modern smartphones focuses on application-layer protections:

- end-to-end encryption
- message integrity
- identity verification
- forward secrecy
- endpoint trust

These are necessary and well-understood controls.

However, many secure messaging systems continue to inherit trust from the underlying mobile communications stack. In practice, this means message content may be protected while the transport path still exposes metadata, signaling behavior, downgrade risk, or radio-layer attack surfaces outside the application's control.

This paper examines an alternative architectural response: remove cellular transport from the trust boundary entirely.

Stingray is not intended as a universal replacement for mainstream messaging systems. It is a transport-constrained communications model for environments in which the carrier path cannot be assumed sufficiently trustworthy for the intended use case.

The contribution of this work is not a new primitive. The contribution is an architectural position implemented in software:

- no trusted dependence on the cellular radio
- fail-closed transport discipline

- local-only key custody
- opaque relay design
- metadata minimization through dependency reduction

2. PROBLEM STATEMENT

The mobile communications stack still carries legacy trust assumptions that are poorly aligned with high-assurance privacy requirements.

Contemporary secure messaging applications often assume that:

- carrier-routed transport remains acceptable if content is encrypted
- phone-number identity is operationally tolerable
- SMS-assisted onboarding or recovery is a manageable tradeoff
- fallback behavior is acceptable if it improves usability
- metadata leakage is secondary to content confidentiality

These assumptions are often reasonable for mass-market messaging.

They are less acceptable in environments where:

- the radio path itself may be monitored or manipulated
- signaling behavior is operationally sensitive
- carrier-linked identity is undesirable
- downgrade or fallback behavior widens exposure
- the application cannot meaningfully harden the inherited transport

In these cases, the problem is not merely weak cryptography. The problem is the retention of trust in a transport model that may itself be the attack surface.

The design question becomes:

Can a messaging system improve security not by endlessly compensating for an inherited mobile transport model, but by refusing to use that transport when it cannot be trusted?

Stingray is one answer to that question.

3. DESIGN POSITION

Stingray is based on a simple design position:

If a transport path is inside the threat model, it should not remain inside the trusted transport set by default.

This yields six architectural rules:

1. Cellular transport is excluded from the trusted transport set.
2. Communication fails closed when only cellular transport is available.
3. Identity keys are generated and stored locally.
4. Relay infrastructure is treated as untrusted storage and delivery plumbing.
5. Metadata minimization is pursued by removing dependency classes, not merely by hiding fields.
6. No server-side recovery path is allowed if it requires server-held decryption material.

This is a deliberately opinionated design. It gives up reach, recovery convenience, and mainstream interoperability in exchange for a narrower and more explicit trust boundary.

4. THREAT MODEL

4.1 Threats considered

Stingray is designed to reduce exposure to the following classes of risk:

- radio-layer interception
- cellular tower impersonation
- forced downgrade behavior on the carrier path
- metadata harvesting inherent to carrier-routed messaging
- passive network observation on permitted non-cellular transport
- relay compromise or curious relay operation
- opportunistic forensic access to powered-off local device state

4.2 Threats not addressed

Stingray does not claim to defend against:

- a compromised endpoint with malware already executing at privileged levels
- malicious baseband, firmware, or hardware implants
- physical device seizure with live unlocked access
- screenshots, photographs of the screen, or user-originated leaks
- compelled disclosure of passphrases or keys
- highly resourced adversaries with deep endpoint compromise

4.3 Boundary statement

Stingray is a bounded transport- and messaging-layer architecture. It does not solve endpoint compromise as a class of problem. It narrows one specific architectural exposure: inherited trust in the cellular communications path.

5. ARCHITECTURE

Stingray is organized as a four-layer architecture.

5.1 Layer 1: Transport discipline

Invariant:

The system does not send, receive, or negotiate keys when the only available path is the cellular radio.

Operational consequences:

- Wi-Fi and Ethernet are allowed
- cellular transport is refused
- no fallback is permitted
- failure is explicit rather than silent

5.2 Layer 2: End-to-end cryptographic envelopes

Once on an allowed transport, messages are protected using:

- X25519 key agreement
- XSalsa20-Poly1305 sealed boxes
- per-envelope ephemeral keypairs
- SAS verification for active MITM resistance at key exchange

5.3 Layer 3: Opaque relay

The relay is designed as a blind mailbox.

Relay-visible fields are limited to:

- recipient public key
- ciphertext
- ephemeral public key
- bucket
- timestamp

The relay has:

- no plaintext visibility
- no account-based sender identity
- no server-side decryption capability
- no recovery material for user vaults

5.4 Layer 4: Local-only identity and storage

Identity keys and message history are generated and stored locally in an Argon2id-protected vault.

This implies:

- no server-side account recovery
- no cloud-held key escrow
- stronger local secrecy when the device is powered off and locked
- higher user burden if keys or passphrases are lost

6. SECURITY PROPERTIES

Subject to the stated threat model, Stingray is intended to provide:

- message confidentiality against relay-side observation
- message confidentiality against radio-layer interception by excluding cellular transport
- reduced metadata exposure relative to carrier-bound identity models
- active-MITM resistance during out-of-band SAS verification
- local key custody without server-held recovery material
- fail-closed behavior when transport guarantees are absent

Stingray does not guarantee:

- anonymity in all network environments
- resistance to compromised endpoints
- safety under all physical access conditions
- secrecy against a fully compromised device

These are deliberate non-claims.

7. COMPARISON WITH EXISTING MESSAGING MODELS

The distinguishing property of Stingray is not that it encrypts messages. Many existing systems do that well.

Its distinguishing property is transport abstention from cellular radio.

Architecturally, the difference can be summarized as follows:

- mainstream secure messengers generally protect content while still permitting carrier-routed mobile transport
- Stingray protects content only after transport has been constrained to non-cellular paths

See whitepaper appendices for comparison table and diagrams.

8. LIMITATIONS

8.1 Usability costs

- no cellular fallback reduces availability
- no recovery makes user mistakes more expensive
- out-of-band verification increases user burden
- strict transport discipline may frustrate non-expert users

8.2 Operational constraints

- the system is not suitable for users requiring always-on reachability over conventional carrier infrastructure
- the system does not solve endpoint malware
- the system does not remove all metadata
- the system depends on users understanding and respecting the transport model

8.3 Evaluation status

At the time of writing, the system remains alpha software.

It has not yet completed:

- formal external cryptographic review
- production-hardware validation

- full end-to-end testing across realistic device conditions
- broader peer review of threat model and usability tradeoffs

Any use beyond research and evaluation would be premature.

9. FUTURE WORK

Planned next-stage work includes:

- optional onion transport
- improved relay unlinkability
- stronger metadata minimization
- self-host relay deployment tooling
- local vault hardening
- hardware-backed key storage integration where practical
- improved SAS verification UX
- external cryptographic review
- broader threat-model challenge and peer critique

These are additive improvements. They do not change the primary design principle: the cellular path remains outside the trusted transport boundary.

10. CONCLUSION

Stingray is built around a narrow but important proposition:

If the transport path cannot be trusted, application-layer encryption alone is not a complete answer.

The system therefore chooses not to compensate indefinitely for inherited mobile transport assumptions. It chooses to remove those assumptions where possible.

This makes the design less convenient than mainstream messaging. It also makes its trust boundary clearer.

That clarity is the point.

The broader lesson is not that every secure messenger should adopt the same model. The broader lesson is that communications architecture should be willing to reject inherited infrastructure assumptions when those assumptions no longer match the threat environment.

In that sense, Stingray is both a software system and an architectural argument: sometimes the most effective hardening step is not to add another protective layer, but to stop using the weakest layer in the first place.

11. DISCLOSURE AND STATUS

This paper describes a research system in alpha state.

It is published for:

- technical review
- architecture critique
- threat-model challenge
- implementation scrutiny

It is not a production-safety claim.

It is an invitation to examine whether transport abstention from cellular radio is a viable design pattern for higher-assurance messaging systems.

KEYWORDS

secure messaging, transport security, mobile security, IMSI catcher,

metadata minimization, end-to-end encryption

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END

Whitepaper Appendices

These appendices are intended to be included after the main paper body. They are separated from the core narrative so the main paper remains readable when rendered into PDF by tools that handle monospace layouts inconsistently

Appendix A. Architectural Comparison

Table A1. High-Level Architectural Comparison

```text

| System   | Cellular Permitted?                  | Phone Number Model? | SMS Bootstrap / Fallback?         |
|----------|--------------------------------------|---------------------|-----------------------------------|
| Stingray | No                                   | No                  | No                                |
| Signal   | Yes                                  | Yes                 | Commonly tied to carrier identity |
| WhatsApp | Yes                                  | Yes                 | Yes                               |
| Threema  | Usually yes                          | Mixed / optional    | Limited                           |
| Session  | Mobile transport may still be used   | No                  | No                                |
| Briar    | Peer-to-peer and resilient by design | No                  | No                                |

```

```text

| System   | Opaque Relay / Minimal Relay State? | Local Key Custody?         | Fail Closed if Only Cellular? |
|----------|-------------------------------------|----------------------------|-------------------------------|
| Stingray | Yes                                 | Yes                        | Yes                           |
| Signal   | Partial                             | Mostly local               | No                            |
| WhatsApp | Partial                             | Platform/account dependent | No                            |
| Threema  | Partial                             | Mostly local               | No                            |
| Session  | Strong metadata resistance          | Local                      | No                            |
| Briar    | Strong local/P2P                    | Local                      | No                            |

```
| | model | | |
+-----+-----+-----+-----+
```
```

Notes:

1. This comparison is architectural and high-level. It is not a claim of superiority across all dimensions.
2. The distinguishing property of Stingray is transport abstention from the cellular radio, not merely end-to-end encryption.
3. Third-party systems should be checked against current official vendor documentation before formal publication.

Appendix B. Diagram Blocks

Diagram B1. Trust-Boundary Shift

```
```text
```

Conventional secure messenger

User

|

v

Application encryption

|

v

Cellular / carrier path

|

v

Relay / service infrastructure

|

v

Recipient

Result:

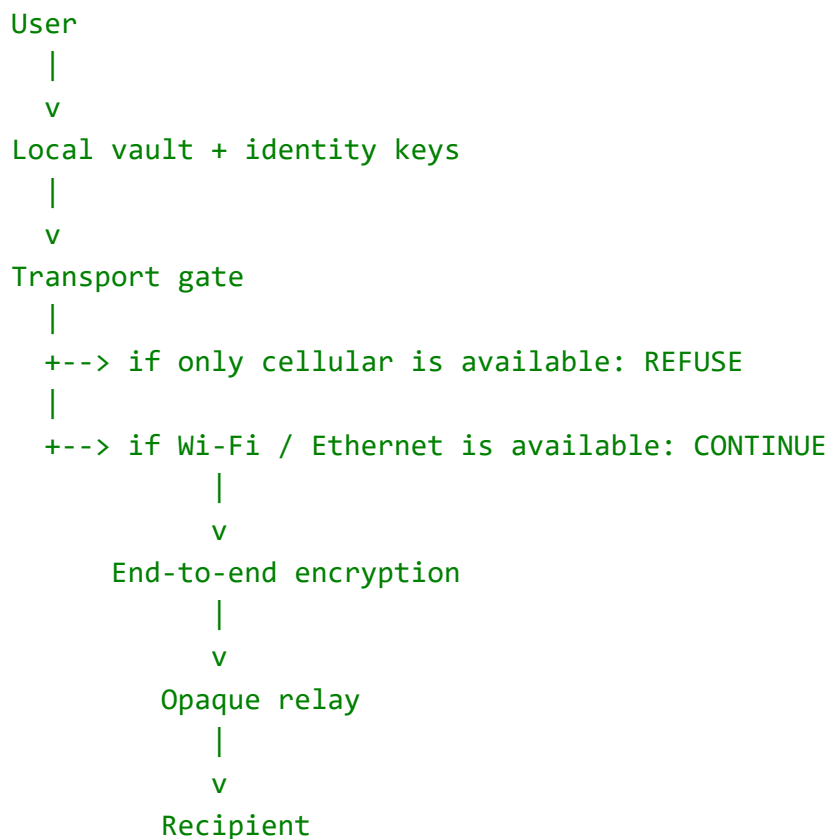
Content may be protected, but the carrier path remains inside the effective operational trust boundary.

```
```
```

```
```text
```

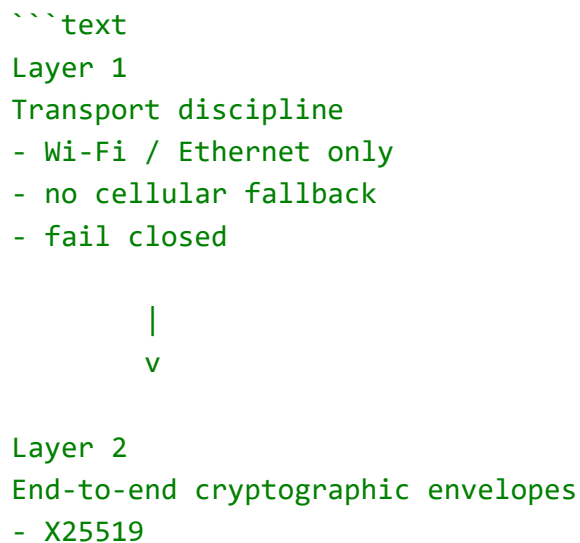
Stingray model





Result:  
Cellular transport is removed from the trusted transport set.  
```

Diagram B2. Layered Architecture



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- XSalsa20-Poly1305
 - ephemeral keypairs
 - SAS verification
- ```

```text

|  
v

Layer 3

Opaque relay

- blind mailbox
- ciphertext only
- minimal routing fields
- no plaintext visibility

|  
v

Layer 4

Local-only identity and storage

- Argon2id-protected vault
  - local keys
  - local history
  - no server-side recovery
- ```

### ### Diagram B3. Message Flow

```text

Sender device

- |
- | 1. verify allowed transport
- | 2. load local identity from vault
- | 3. generate ephemeral envelope key
- | 4. encrypt message

v

Opaque relay

- |
- | stores only:
- | recipient_pubkey

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```
| ciphertext
```
```

```
```text
| ephemeral_pubkey
| bucket
| timestamp
v
Recipient device
|
| 1. fetch envelope
| 2. decrypt locally
| 3. persist locally
v
Readable message
```

```
No server-side plaintext.
No server-side recovery material.
No cellular fallback path.
```
```

## ## Appendix C. Suggested Placement

- Place ``Table A1`` under a section titled ``Appendix A. Comparative Notes``.
- Place ``Diagram B1`` immediately after the architectural comparison section the PDF renderer preserves monospace formatting cleanly.
- Otherwise move all diagrams into ``Appendix B. Diagram Blocks``.
- If the main PDF renderer keeps breaking wide tables, export these appendices as separate pages or include them as image plates in the final PDF.